Adapted from QUBE-SERVO 2 Workbook – Student, © 2021 Quanser Inc.

Hardware Interfacing, Filtering, and Modelling

Source and modifications. This manual is an adaptation of three Quanser student workbooks for the QUBE-Servo 2 — *Hardware Interfacing*, *Filtering* and *Block Diagram Modeling* (QUBE-SERVO 2 Workbook – Student, v1.2, © 2021 Quanser Inc.) — combined into a single laboratory experiment and used under the licence reproduced at the end of this document.

Retained from the original workbooks: the encoder, DC motor and low-pass filter background; the motor armature model, Figures 1 to 4 and the parameter values in Table 1; the QUARC configuration, encoder-reading and motor-driving procedures in Sections 6.1 to 6.3; the filtering exercises in Section 6.4; and the block-diagram and transfer-function modelling exercises in Sections 6.6 and 6.7.

Modified from the original: the QUARC configuration steps have been updated to the current Simulink *Hardware* tab workflow; the Stall Detection subsystem (Figure 4) has been modified, with the stall threshold changed from 20 s to 10 s; and the cut-off frequency sweep in Section 6.4 has been removed, since filter design is covered later in the course. Explanatory notes on pre-lab preparation, solver selection and reuse of the Lab 1 Euler function have been added.

Written for ENME 561/661 by the course instructor: the required materials, end-of-session shutdown procedure and deliverables; the differential-equation modelling section (Section 6.5), which has no counterpart in the original workbooks; the three-model comparison and analysis in Section 6.8; and the whole of Section 7, including Table 2 and the peer design review.

Table of Contents

Hardware Interfacing, Filtering, and Modelling 1

1 Required Material 3

2 End-of-Session Shutdown Procedure 3

3 Objective 3

4 Deliverables 3

5 Background 3

5.1 DC Motor 3

5.2 Encoders 4

5.3 Filtering 4

5.4 Motor Modeling 4

6 In-Lab Exercises 6

6.1 Configuring a Simulink Code for the QUBE-Servo 2 6

6.2 Reading the Encoder 6

6.3 Driving the DC Motor 7

6.4 Filtering 8

6.5 Modelling using Differential Equations 8

6.6 Modelling using Block Diagram 8

6.7 Modelling using the Transfer Function 9

6.8 Comparison 9

7 Design of Experiments for Parameter Identification of a DC motor 10

7.1 Design Requirements and Constraints 10

7.2 Experiment Design 10

7.3 Peer Design Review 11

1 Required Material

- Hardware: QUBE-Servo 2, Cables, Laptop
- Software: Stall detection block (available on D2L)
- QUBE-Servo 2 User Manual.pdf (available on D2L)

2 End-of-Session Shutdown Procedure

At the end of each lab session, please ensure the following steps are completed:

1. Make sure the QUARC controller is stopped.
2. Turn off QUBE-Servo 2.
3. Shut down the laptop.
4. Place the laptop in the designated cabinet.

3 Objective

In this lab, you work with a DC motor and a rotary encoder. You measure the shaft angle and speed from raw encoder counts, and you implement a first-order low-pass filter to reduce the noise in the speed measurement. You model the DC motor using three techniques, differential equations, block diagrams, and transfer functions, and you compare the behavior of these models with that of the physical motor. You then design an experiment to identify the motor resistance and the back-emf constant, and you review another team's design.

4 Deliverables

- **Section on Modelling using Differential Equations**: Submit one **.mlx file** and its corresponding results in a **PDF**.
- **Rest of Section 6**: Submit one **.slx file** with the **results and discussion** included in the **main PDF**.
- **Section 7**: Submit as a **separate PDF file** by the Lab 2 deadline, and bring a physical copy of the file to the 1st Lab 3 session.

In total: Three **PDF files**, one **.mlx file**, and one **.slx file**.

> **Answering the questions.** Questions marked *(For your own understanding — no written answer required)* are there to direct your attention while you work at the bench; they are not graded and need not appear in your report. Every other question in this manual must be answered in the submitted report.

5 Background

5.1 DC Motor

Direct-current (DC) motors are one of the most common actuators and are used in a variety of applications. The QUBE-Servo 2, shown in Figure 1, has a brushed DC motor that is connected to a PWM amplifier.



Figure 1. QUBE-Servo 2

5.2 Encoders

An encoder is a measurement device that is used to find the position of a spinning shaft. There are many types of encoders, and one of the most common is the rotary incremental optical encoder, shown in Figure 2. Unlike potentiometers, encoders come in two different types: absolute or relative. Incremental encoders are relative. The angle they measure depends on the last position and when it was last powered.

The zeros and ones output of photo sensors attached to the disk, combined with a decoding algorithm, result in a count-based output giving the rotational position of the shaft. This count can then be converted into either radians or degrees based on how many counts there are in a revolution of the shaft.



Figure 2. Incremental rotary optical shaft encoder from US Digital Motion Control Products

5.3 Filtering

A low-pass filter can be used to block out the high frequency components of a signal. A first order low-pass filter transfer function has the form

$$G(s) = \frac{\omega_f}{s + \omega_f}, \tag{1.1}$$

where ω_f is the cut-off frequency of the filter in radians per second (rad/sec).

5.4 Motor Modeling

The Quanser QUBE-Servo 2 is a DC rotary servo system. Its motor armature circuit schematic is shown in Figure 3 and the values of electrical and mechanical parameters are given in Table 1. The DC motor shaft is connected to the load hub. The hub is a metal disk used to mount the red disk and has a moment of inertia of J_h. A disk load is attached to the output shaft with a moment of inertia of J_d.

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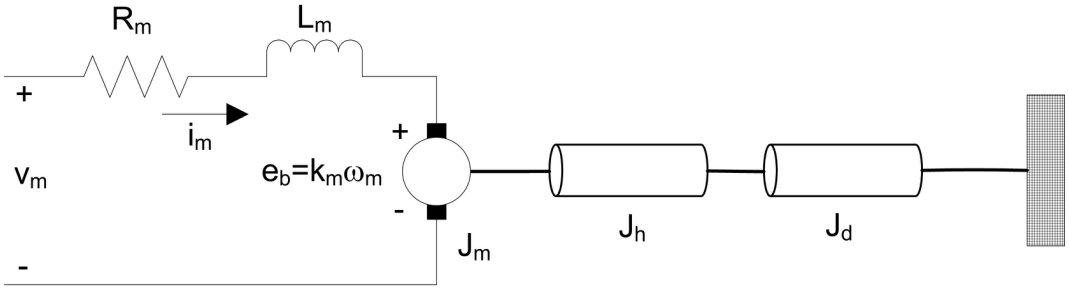


Figure 3. QUBE-Servo 2 DC motor and load

Table 1: QUBE-Servo 2 system parameters

Symbol	Description	Value
DC Motor		
R_m	Terminal resistance	8.4Ω
k_t	Torque constant	0.042 N.m/A
km	Motor back-emf constant	0.042 V/(rad/s)
Jm	Rotor inertia	4.0×10^{-6} kg.m^2
Lm	Rotor inductance	1.16 mH
m_h	Load hub mass	0.0106 kg
rh	Load hub radius	0.0111 m
J_h	Load hub inertia	0.6×10^{-6} kg.m^2
Load Disk		
m_d	Mass of disk load	0.053 kg
r_d	Radius of disk load	0.0248 m

The back-emf (electromotive) voltage, shown by $e_b(t)$ depends on the speed of the motor shaft, ω_m, and the back-emf constant of the motor, shown by k_m. It opposes the current flow. The back emf voltage is given by:

$$e_b(t) = k_m\, \omega_m(t). \tag{1}$$

Using Kirchhoff's Voltage Law, we can write the following equation:

$$v_m(t) - R_m i_m(t) - L_m \frac{di_m(t)}{dt} - k_m \omega_m(t) = 0. \tag{2}$$

Since the motor inductance L_m is much less than its resistance, it can be ignored. Then, the equation becomes

$$v_m(t) - R_m i_m(t) - k_m \omega_m(t) = 0. \quad (3)$$

Solving for $i_m(t)$, the motor current can be found as:

$$i_m(t) = \frac{v_m(t) - k_m \omega_m(t)}{R_m}. \quad (4)$$

The motor shaft equation is expressed as

$$J_{eq} \frac{d\omega_m(t)}{dt} = \tau_m(t), \quad (5)$$

where J_{eq} is the total moment of inertia acting on the motor shaft and τ_m is the applied torque from the DC motor. Based on the current applied, the torque is

$$\tau_m = k_t i_m(t), \quad (6)$$

where k_t is the motor current torque constant.

6 In-Lab Exercises

In this lab, we will be investigating how to work with the QUBE-servo 2 system, how to filter the output to remove noise and look at a couple of different ways to model the system and how they compare with the real system.

Make sure that the QUBE-Servo 2 is connected to the computer via a USB cable and is connected to power using the power cable labelled “Qube Only”. The power LED and USB cable LED need to be green, and since the system is not running properly, the status LED is red.

Pre-Lab. Section 6.5 (Modelling using Differential Equations) does not require the QUBE-Servo 2. Derive the differential equations and draft the corresponding .mlx Live Script *before* your lab session. The scheduled lab period does not allow enough time to complete the whole lab if this section is left until then.

6.1 Configuring a Simulink Code for the QUBE-Servo 2

Follow these steps to build a SIMULINK code that will interface to the QUBE-Servo 2 using QUARC:

1. Load the MATLAB software.
2. Create a new SIMULINK file.
3. Insert the **HIL Initialize block**. This block is used to configure your data acquisition device.
4. Double click on the HIL Initialize block, and in the *Board type* field, select ***qube_servo2_usb***.
5. Under the QUARC tab, select ***quarc_win64.tlc*** as the QUARC Target. After this, you should have a new tab called Hardware.
6. In the Hardware tab, click **Monitor & Tune**. If your system has been successfully connected, the status LED on the Qube Servo will turn green.
7. Clicking the **Stop** button in the Hardware tab will now stop the program running on the system.

6.2 Reading the Encoder

1. Add the **HIL Read Encoder** block.
2. Connect the HIL Read Encoder to a **Display block** (without the HIL Write Analog block).

3. Click on **Monitor and Run**
4. Manually **rotate** the red disc back and forth. The Display block shows the number of counts measured by the encoder. The encoder counts are proportional to the angle of the disc.
5. What happens to the encoder reading every time the QUARC controller is started? Stop the controller, move around the disc, and re-start the controller. What do you notice about the encoder measurement when the controller is re-started? *(For your own understanding — no written answer required.)*
6. Measure how many counts the encoder outputs for a full rotation. Validate that this matches the specifications given in the QUBE-Servo 2 User Manual.
7. Ultimately, we want to display the disc angle in degrees, not counts. Modify your code to convert counts to degrees. Run the QUARC controller and confirm that the Display block shows the angle of the disc correctly. What conversion factor did you use? Add a screenshot of the code that does this calculation
8. If instead, you wanted to measure your angle in radians, what conversion factor would you use? Make this modification to your system and confirm the display block is showing the expected result.

6.3 Driving the DC Motor

1. Add the **HIL Write Analog** block. This block is used to output a signal from the analog output channel #0 on the data acquisition device. This is connected to the on-board PWM amplifier, which drives the DC motor.
2. Add the **Constant block** and connect the Constant block to the HIL Write Analog blocks.
3. Add the HIL Read Analog blocks, if you have not done so.
4. Add the **Display block** and connect the Display block to the HIL Read Analog blocks, if you have not done so.
5. Insert the **Stall detection block** found on D2L into your system. The **control input** to this block is the signal before the HIL write block, so it is reading the voltage you are sending to the motor. The **position (counts) input** should be connected right after the HIL Read Encoder block.
   This subsystem will stop the code if the shaft is not moving for more than 10 seconds and more than 5 volts are applied. The details of this subsystem are shown in Figure 4.

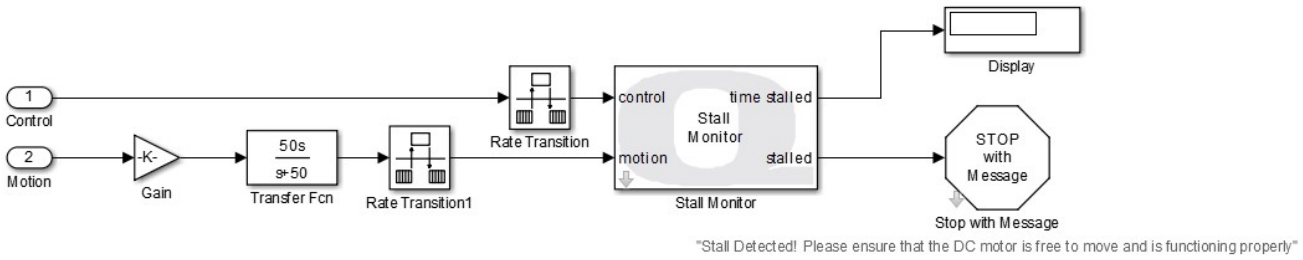


Figure 4: Stall Detection Block

6. **Build and Run** the QUARC controller.
7. Set the Constant block to 0.5. This applies 0.5 V to the DC motor in the QUBE-Servo 2.
8. Confirm that we are obtaining a *positive measurement when a positive signal is applied*. This convention is important, especially in control systems, when the design assumes the measurement goes up positively when a positive input is applied.
9. In what direction does the disc rotate (clockwise or counter-clockwise) when a positive input is applied? *(For your own understanding — no written answer required.)*

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10. **Stop** the QUARC controller.

6.4 Filtering

1. Now, modify your code so that you can measure the **speed** that the motor is running at, and add a **Scope** block, then output that information to the Scope. Additionally, change the input from the constant value to a **square** wave going from 1V to 3V at 0.4 Hz.
2. Build and run the QUARC controller. Examine the encoder speed response. How do the input and output signals compare? Add the plots showing both signals on one figure to your report.
3. Explain why the encoder- based measurement is noisy. **Hint:** Measure the encoder position measurement using a new Scope. Zoom up on the position response and remember that this later enters a derivative block. Is the signal continuous? Sow both position and speed plots in your report.
4. One way to remove some of the high- frequency components is to add a low-pass filter (LPF) to the derivative output. Add a **Transfer Fcn block** after the Derivative output and connect LPF to the Scope. Set the Transfer Fcn block to act as the following transfer function $50/(s+50)$.
5. Build and run the QUARC controller. Show the filtered encoder-based speed response and the motor voltage. Has it improved? *(For your own understanding — no written answer required.)*
6. What is the cutoff frequency of the low-pass filter $50/(s+50)$? Give your answer in both rad/s and Hz.
7. **Stop** the QUARC controller.

6.5 Modelling using Differential Equations

Deliverables: MATLAB Live Script (.mlx)[1], its PDF document that includes: The derivation of the governing differential equations and the plots of shaft rotational angle position, speed, and input voltage over time.

This section is to model the DC motor using Differential Equations and numerically solve the model using MATLAB Script.

1. Using Newton's Law, the Voltage Law, and the Motor Modelling section of this manual, derive the set of differential equations that model the shaft's rotational angle position with voltage as the input. The unknown in the differential equation is the angular position (θ), and the input is the voltage.
2. Next, create a live script in the Live Editor (i.e., an `.mlx` file) to numerically solve for the shaft's rotational angle position (θ) over the time interval 0 to 10 seconds (e.g., using Euler's method). Assume that no voltage is applied during the first second, and that a constant voltage of 5 V is applied from 1 second onward.
   The output of the code should include plots of the shaft's rotational angle position, and speed over the 10-second interval. You must upload both your `.mlx` code file and a PDF export of the live script that contains the code and the plots of the solution over time with the input voltage.

> **Note.** Reuse the Forward Euler function you wrote in Lab 1 rather than writing a new solver. State in your Live Script which function you reused and describe any changes you made to it.

6.6 Modelling using Block Diagram

Deliverables: Code in the Main Simulink model file (.slx) and a screenshot of the code and discussion of the results included in the main PDF file.

[1] If you are interested to use Simulink to solve the ODE, read https://blogs.mathworks.com/simulink/2008/05/23/how-to-draw-odes-in-simulink/ Please note that this is out of the scope of this lab.

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This section is to model the DC motor using a Block Diagram and numerically solve the model using MATLAB Simulink.

Note on the Simulink solver[2]. Left at its default *Auto* setting, Simulink may select a variable-step solver whose steps do not align with the fixed sample time of the QUARC blocks, and your model output will then differ noticeably from the hardware measurement for reasons that have nothing to do with the model. In Model Settings, under Solver, select a fixed-step solver (*ode1*, Forward Euler) with a step size equal to the sample time used by the HIL blocks. Report the solver and the step size you used. If you observe a difference between the *Auto* result and the fixed-step result, include both and comment on it in the Comparison section.

1. Build a block diagram model of the QUBE-Servo 2 in MATLAB Simulink. Add its screenshot and upload the code to D2L.
2. Connect the block diagram model of the QUBE-Servo 2 in parallel with the SIMULINK code to drive the motor, so the motor and the Simulink model receive the same input voltage command. Build and run the QUARC controller with your QUBE-Servo 2 model.
3. Stop the QUARC controller.

6.7 Modelling using the Transfer Function

Deliverables: Code in the Main Simulink model file (.slx) and a screenshot of the code and derivation of the transfer function, plots of the input and output and a discussion of the results included in the main PDF file.

1. Take the Laplace Transform of the differential equations to model the DC motor (the shaft rotational angle's speed). Find the speed to the voltage transfer function, $\Omega(s)/V_m(s)$, of the system. Evaluate the transfer function by filling in the numerical values for the variables.
2. Connect the block diagram model of the QUBE-Servo 2 in parallel with the SIMULINK code from the previous section to run the motor, so the motor and the Simulink model receive the same input voltage command. Build and run the QUARC controller with your QUBE-Servo 2 model.
3. Stop the QUARC controller.

6.8 Comparison

In this section, you will compare all the different models you created and numerically simulate the system using these models.

- Plot Comparison: Plot the outputs of all three DC motor models and the output of sensors together on a single figure over a 10-second interval.
- Analysis:
    - Compare the similarities and differences among the models.
    - Discuss whether your models represent the QUBE-Servo 2 reasonably well.
    - Identify and explain the causes of mismatches between the models and the experimental system. Provide at least one possible source of discrepancy and explain how it would affect the model results.

[2] https://www.mathworks.com/help/simulink/ug/choose-a-solver.html

7 Design of Experiments for Parameter Identification of a DC motor

Deliverables: Upload your work as a single PDF file to D2L. At the beginning of Lab 3, you must also bring a printed copy and hand it in to your teaching assistant. The TA will provide your printed copy to another group, who will follow the instructions and evaluate your work. **You will in turn receive another group's procedure and complete the peer review form in the Peer Design Review section during the Lab 3 session. Your mark for this section reflects both the quality of your own design and the quality of the review you provide.**

7.1 Design Requirements and Constraints

Before you write any procedure, complete Table 2. An experiment design begins with what you are allowed to do and what you are able to measure, not with the steps themselves. You will not be able to answer every row from this manual alone: the QUBE-Servo 2 User Manual on D2L contains the electrical and mechanical limits you need.

Table 2: Design requirements and constraints

Item	Your entry	Source or justification
Parameters to be identified		
Quantities you will measure, and the block or instrument used for each		
Maximum voltage you will apply, and why		
Stall constraint: maximum voltage and maximum duration for which the shaft may be held stationary		
Any other constraint that protects the hardware (duty cycle, cool-down between trials, current limit)		
Equipment limits that bound your measurement (amplifier range, current-sense range, encoder resolution, sample time)		
Number of trials, and number of repetitions per trial		
Assumptions your procedure relies on		

7.2 Experiment Design

Design one or a series of experiments to identify the following parameters:

- Motor resistance (R_m), and
- Motor back-emf constant (k_m)

Assume that you can measure current, voltage and shaft speed.

Your procedure must permit ***R_m* and *k_m* to be determined independently of one another. A procedure that yields only a combination of the two — for example, one that records shaft speed against applied voltage without ever measuring current — has not identified either parameter, and will be marked accordingly.**

Ensure that your instructions for the experiments are clear and detailed enough for another student to follow and implement.

7.3 Peer Design Review

At the beginning of Lab 3 your teaching assistant will exchange the printed procedures between teams. You will have approximately 25 minutes to read and evaluate the procedure you receive, using the peer review form provided in the Lab 3 manual. For each criterion, assign a rating and write one or two sentences justifying it. A rating without a justification earns no marks. Return the completed form to your teaching assistant before the end of the session; a copy will be given to the team whose work you reviewed.

The form carries five criteria, worth two points each. Design your own procedure with these in mind:

- **Clarity of experiment** — could another team execute your procedure exactly as written, without asking you a question?
- **Parameter identification** — does the procedure determine the two parameters independently, rather than yielding only a combination of them?
- **Experimental setup** — is the setup practical, safe and logically organized, with voltage and stall limits stated?
- **Data collection and method** — is it clear how each quantity is measured and recorded, with repetitions and an estimate of uncertainty?
- **Feasibility and realism** — is the design achievable with the available hardware and instrumentation?

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